ROUND 1 · CHART-GENERATION BENCHMARK

13 songs. 3 generators. 195 charts. One question.

We built an AI step-chart generator. Before believing our own press, we ran the same 13 songs through the tools that already exist — the one the community actually uses, and the one the academics actually cite — and scored all three with the identical scorer, on the identical audio.

ROUND 1 → **STEPFORGE** wins on timing and danceability · loses on dynamics

THE CONTENDERS

Who's in the ring

OURS

STEPFORGE

DSP onset inventory → a language model that may only reference onset IDs, never timestamps → a foot-flow cost solver → hard gates.

THE PEOPLE'S CHOICE

AutoStepper

Java, 2018, pure DSP and rule-based patterns. What the community has actually used for eight years. The bar you must clear to be worth existing.

THE ACADEMIC

DDC

Dance Dance Convolution, ICML 2017. CNN→LSTM onset placement plus a conditional LSTM step selector. Run from the author's original checkpoints.

READ THIS FIRST

Two of our metrics are rigged in our favour

DISCARDED — TAUTOLOGICAL

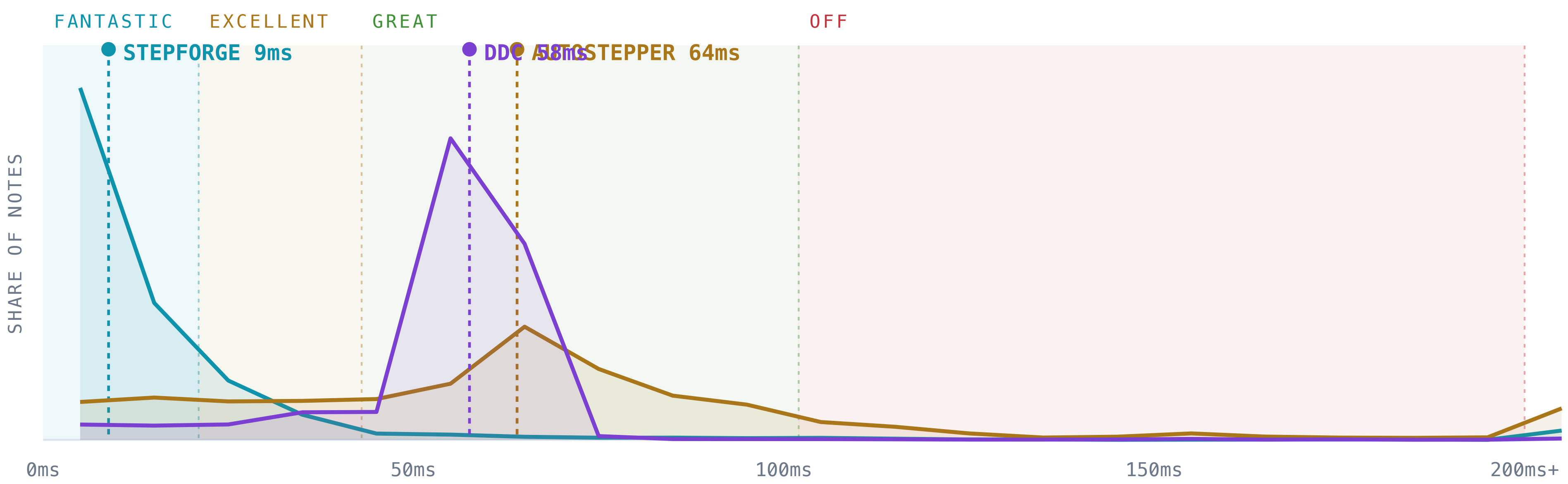
Onset alignment and **BPM error** are scored against our own DSP analysis. Our designer places notes *by referencing IDs from that exact onset inventory*, and our simfile's BPM *is* the BPM that analysis measured. Of course we score 0.94 and 0.000. That is a tautology, not a victory. Both rows are thrown out.

What's left are the questions our own onset detector cannot rig: **in absolute milliseconds, where do the notes actually land? — is the chart physically comfortable to dance?** (pure arrow geometry; it never touches the audio) — and **does the chart's density follow the song's energy?**

FINDING 01 · TIMING

Where the notes actually land

Every note of all 195 charts, measured against the audio's real transients, then bucketed by **ITGmania's own judgment windows** — the game's definition of "on time", not a threshold we invented.



Distribution of absolute timing error. The bands are the real ITG judgment windows. The vertical tick on each curve is that generator's median note.

STEPFORGE MEDIAN

8ms

81.4% of notes land inside ±21ms — a **Fantastic**.

AUTOSTEPPER MEDIAN

64ms

Only **13.8%** Fantastic. It detects tempo per song, and still misses.

DDC MEDIAN

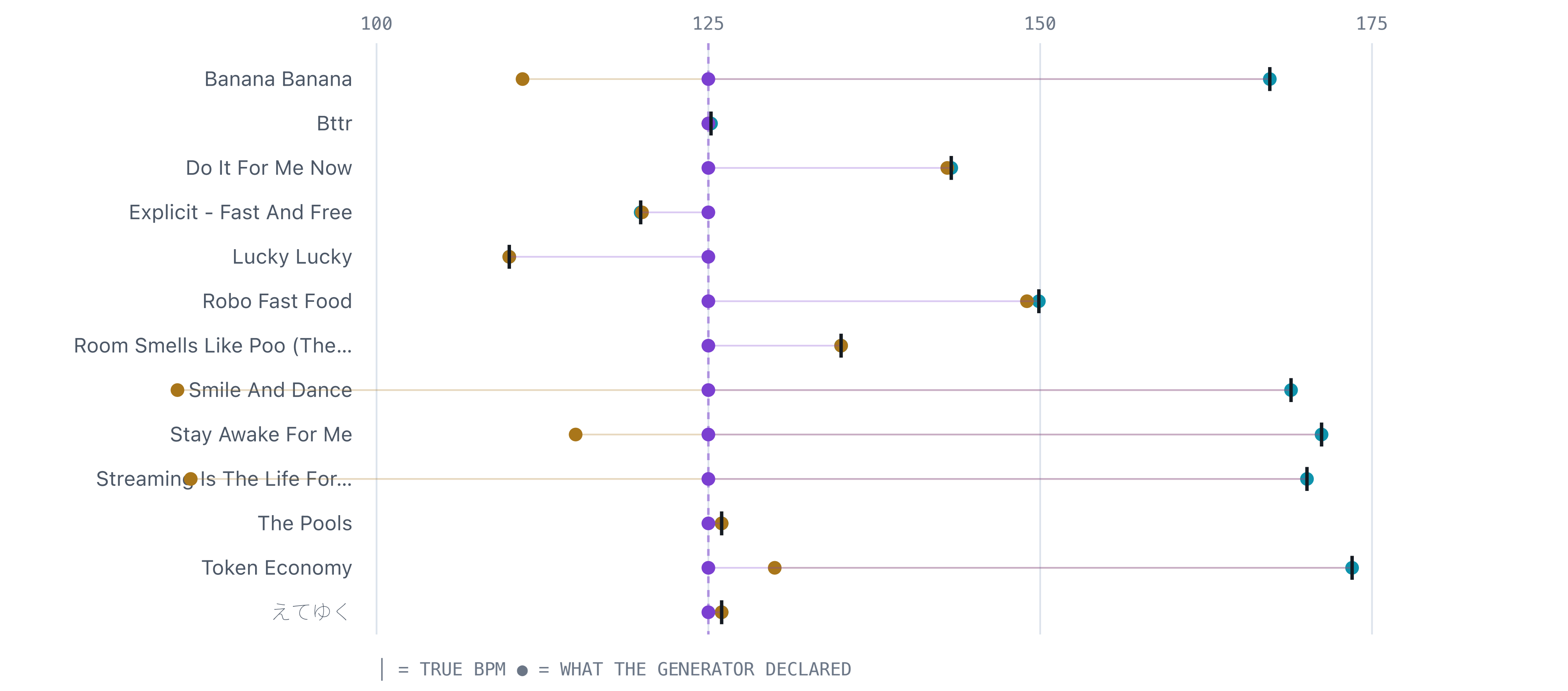
58ms

5.0% Fantastic — but **99.1%** within ±102ms. It hears the music. Keep reading.

FINDING 02 · TEMPO

DDC thinks every song is 125 BPM

Not approximately. Literally `125.0` for all thirteen — whether the track is 110 or 173.5.



Declared BPM (what the simfile says) versus true BPM (what the audio is). Each song is a row. DDC's marks form a perfect vertical line at 125.

WHY DDC'S 99.1% MATTERS MORE THAN ITS 5%

DDC is *not* placing notes randomly — 99% of them are within ±102ms of a real transient. **It hears the music perfectly well. It just cannot write down *when*.** It has no tempo model: it detects onsets in continuous time and then snaps them onto a fake 125 BPM grid. That grid's resolution *is* the ~58ms error, which is why DDC's median is almost identical on every song regardless of its actual tempo.

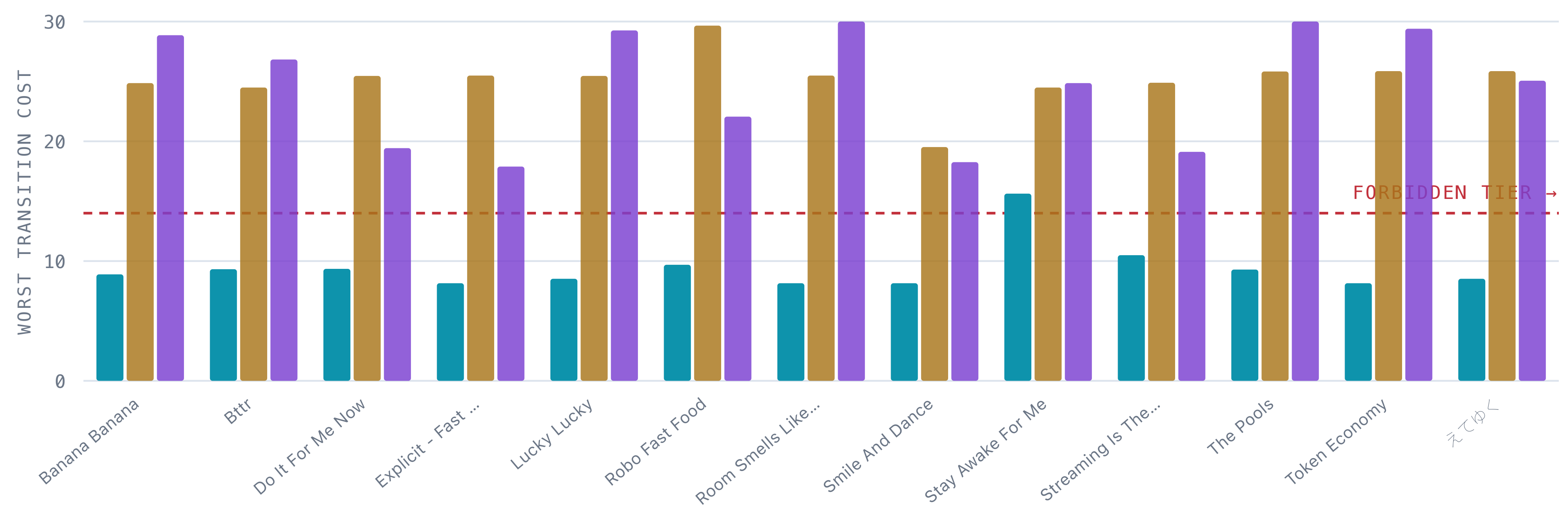
A note 58ms late is not a Fantastic, or even an Excellent. In ITG it is a Great — the judgment you get when you are visibly off the beat. **A bot playing DDC's own chart, perfectly, would be graded mediocre.**

That an independently-trained 2017 CNN agrees with our onset detector about *where* the music is — while disagreeing about precision — is also the best evidence that our onset ground truth is sound.

FINDING 03 · DANCEABILITY

The metric we cannot rig

Foot-flow cost is a state machine over (left foot, right foot, who moved last). It penalises double-steps, jacks and crossovers, and rewards clean alternation. **It never looks at the audio** — no onset detector can flatter it.



Worst transition in each chart, averaged per song. Above the dashed line are forbidden-tier patterns: moves that force a double-step or a physically awkward spin.

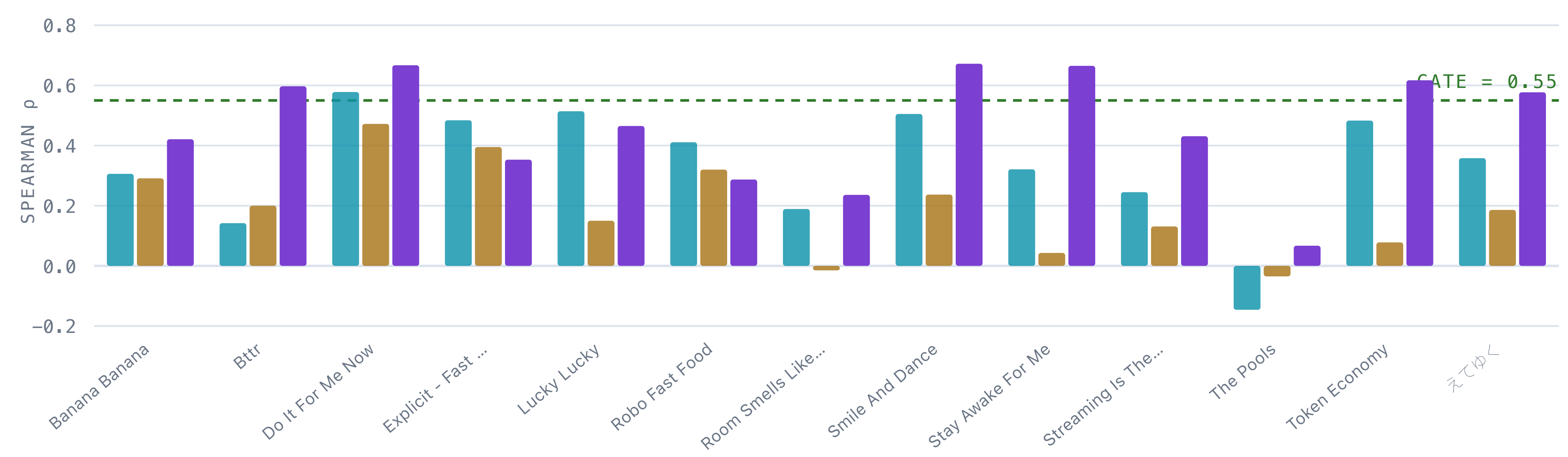
THE REAL RESULT

AutoStepper passes the flow gate on zero of its 65 charts. DDC passes on 8. STEPFORGE passes on 63. Both baselines can find the beat, roughly. Neither has any model of the fact that a human has two feet.

FINDING 04 · DYNAMICS

Where we lose — and we do lose

Spearman correlation of note density against the song's energy curve. Does the chart get busier when the *music* gets busier?



Density-vs-energy correlation per song. Higher is better. DDC is ahead on 10 of 13.

DDC BEATS US, AND IT DESERVES TO

DDC's placement network is conditioned on difficulty and was trained on thousands of human-authored charts — so it *learned* what human charters do: go quiet in the breakdown, go wild in the drop. Our designer is merely *told* to in a prompt, and manages it about half as well: **ρ 0.338** against DDC's **0.466**, passing the gate 23% of the time to DDC's 43%.

Our charts are precisely placed and comfortable to dance, but flatter in their dynamics than a model from 2017. That is the most useful thing this benchmark produced, and it is a to-do, not a footnote.

ALL THIRTEEN

Song by song

Median timing error per song. The bar is capped at 100ms; the ITG "Fantastic" window (21ms) is marked.

Banana Banana
167.3 BPM — 989 onsets
9ms
38ms
59ms

Bttr
125.2 BPM — 794 onsets
8ms
76ms
59ms

Do It For Me Now
143.3 BPM — 913 onsets
10ms
57ms
57ms

Explicit - Fast And Free
119.9 BPM — 444 onsets
6ms
64ms
60ms

Lucky Lucky
110 BPM — 531 onsets
7ms
64ms
59ms

Robo Fast Food
149.9 BPM — 566 onsets
8ms
85ms
62ms

Room Smells Like Poo (The Dog Just Took A Crap On The Floor)
135 BPM — 880 onsets
5ms
62ms
56ms

Smile And Dance
168.9 BPM — 917 onsets
3ms
34ms
57ms

Stay Awake For Me
171.2 BPM — 1601 onsets
5ms
39ms
53ms

Streaming Is The Life For Me
170.1 BPM — 873 onsets
4ms
43ms
57ms

The Pools
126 BPM — 1337 onsets
19ms
67ms
59ms

Token Economy
173.5 BPM — 1190 onsets
17ms
75ms
58ms

えてゆく
126 BPM — 1038 onsets
10ms
86ms
57ms

THE NUMBERS

Full table

SONG	BPM	SF ERR	AS ERR	DDC ERR	SF FLOW	AS FLOW	DDC FLOW	SF P	AS P	DDC P
Banana Banana	167.3	9	38	59	8.9	24.9	28.9	0.31	0.29	0.42
Bttr	125.2	8	76	59	9.3	24.5	26.8	0.14	0.20	0.60
Do It For Me Now	143.3	10	57	57	9.3	25.5	19.4	0.58	0.47	0.67
Explicit - Fast And Free	119.9	6	64	60	8.2	25.5	17.9	0.48	0.40	0.35
Lucky Lucky	110	7	64	59	8.5	25.5	29.3	0.51	0.15	0.47
Robo Fast Food	149.9	8	85	62	9.7	29.7	22.1	0.41	0.32	0.29

Room Smells Like Poo (The Dog Just Took A Crap On The Floor)	135	5	62	56	8.2	25.5	30.2	0.19	−0.01	0.24
Smile And Dance	168.9	3	34	57	8.2	19.5	18.3	0.51	0.24	0.67
Stay Awake For Me	171.2	5	39	53	15.6	24.5	24.9	0.32	0.04	0.67
Streaming Is The Life For Me	170.1	4	43	57	10.5	24.9	19.1	0.24	0.13	0.43
The Pools	126	19	67	59	9.3	25.8	30.4	−0.15	−0.04	0.07
Token Economy	173.5	17	75	58	8.2	25.9	29.4	0.48	0.08	0.62
えてゆく	126	10	86	57	8.5	25.9	25.1	0.36	0.19	0.58

ROUND 1

Who won

> DECISION: STEPFORGE, ON POINTS — NOT A KNOCKOUT
>
> **Won:** timing, by an order of magnitude (8ms against 58 and 64) — the difference between a Fantastic and a Great on every single note. And danceability, which is not close: a 96.9% flow-gate pass against 12.3% and 0%.
>
> **Lost:** dynamics. A model from 2017 shapes a song's energy better than our language-model designer does.
>
> **Thrown out:** onset alignment and BPM accuracy. We graded our own homework on those.

Round 2 is fixing the dynamics gap without giving back the timing or the foot-flow.

METHOD

13 songs · 5 difficulties · 3 generators · 195 charts · 39,483 notes scored. Identical audio for every generator; only the chart differs. Each simfile is rebased into the audio's real time domain before scoring, because every generator's beats are relative to its own tempo map and they disagree wildly. DDC was run from the author's original pretrained checkpoints inside his own Docker image (its native Python 2 / TensorFlow 0.12 stack no longer installs); AutoStepper from its shipped JAR.